



GANESH CHHETRI

Software Engineer

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Portfolio

1. PROFILE

Software engineer with three years of experience developing and supporting enterprise applications in production at NGN Technologies Private Limited, Thimphu. The work covers the full lifecycle: gathering requirements with business users, designing the data model, building the services and screens, deploying to cloud or on-premise infrastructure, and handling support and security work after release. Currently project lead on two systems for government agencies. A recurring part of the work has been integrating Bhutan NDI, the national digital identity, into existing systems. Holds a four-year B.Tech in Computer Science & Engineering with specialisation in Cloud Technology and Information Security (CGPA 8.38, First Division with Distinction), and is an AWS Certified Cloud Practitioner and Certified DevOps Engineer.

2. EMPLOYMENT

Designation	Organization	From	To
Assistant Software Engineer (Grade T1)	NGN Technologies Private Limited e-Centric Division, Thimphu	Aug 2023	Jan 2024
Software Engineer (Grade S1)	NGN Technologies Private Limited e-Centric Division, Thimphu	Feb 2024	Feb 2026
Software Engineer (Grade T2, Band 2)	NGN Technologies Private Limited e-Centric Division, Thimphu	Mar 2026	Present

Three years of continuous experience since August 2023, entirely on production systems.

3. PROJECT EXPERIENCE

Period	Project	Role	Scope and Contribution
2025 – Present	Mines Administration System	Project Lead	A 34-module platform for the Department of Geology and Mines covering mining and quarrying leases, transport permits, stockyard balances, royalty calculation and online payment, multi-level approval workflows and role-based access control. Delivered through user acceptance testing and deployed on Kubernetes.
2025	DRUG-RIMS	Project Lead	Registration and licensing of pharmaceutical products for the Bhutan Food & Drug Authority, including a Certificate of Product Registration rebuilt to an official document standard. A security hardening exercise closed OWASP ZAP findings, fixed an authorization bypass, configured content security policy and headers at the proxy, and cleared flagged dependency vulnerabilities.
2024 – Present	Bhutan NDI integration	Across projects	Bhutan NDI integrated into DRUG-RIMS, the Mines Administration System, DRA, BCSEA and the G2C Feedback System — credential issuance, proof requests against the citizen identity wallet, passwordless login, webhook registration and event handling over NATS.
2024 – 2025	Digital Health Platform	Developer	Application modules for a national health screening programme — dynamic screening, assessment, consent capture, behaviour tracking, email verification and search — with the AWS infrastructure behind them: scheduled Lambda warming, VPC and NAT gateway networking, bulk user provisioning and IAM policy design.
2024	ETool	Developer	Report definition, report generation and the data access layer beneath them.
2024	G2C Feedback System	Support and enhancement	A citizen feedback platform for government-to-citizen services, integrated with Bhutan NDI and using NATS for event handling.

2024	BCSEA examination services	Support and enhancement	Online payment integration for examination services, and optimisation of the views behind the production reporting database.
2024	Food and drug regulatory systems	Support and enhancement	Maintenance and enhancement of live regulatory systems: new features, resolution of long-standing defects, foreign data wrapper links for cross-database reporting, recovery of a medical products database from character-encoding corruption, and correction of CORS configuration across services.
2023 – 2024	Report Management System	Developer	Report definition, generation and export for end users, built in Python.
2023 – 2024	User Management System	Developer	Account creation and lifecycle, role definitions and the permission model determining what each role can access, built in Python.
—	Document and invoice automation	Independent work	Tooling to generate Word documents and invoices for Nimalung Dratshang and Yuvan Infotech, replacing preparation done manually.

4. CORE RESPONSIBILITIES

- **Project delivery** — Lead delivery of two government systems end to end: requirement gathering, technical development plan, development, integration, documentation, user acceptance testing and go-live, coordinating between business units and the technical team.
- **Application maintenance** — Maintain integrated enterprise applications and customer-facing portals in production, covering releases, defect resolution and user support.
- **Application security** — Remediation of OWASP ZAP findings, content security policy and header configuration, closure of an authorization-bypass flaw, dependency vulnerability clearance and proxy hardening.
- **Database administration** — Administration of production PostgreSQL and MySQL databases: backup and restoration, foreign data wrapper links for cross-database reporting, view and query optimisation, and recovery from character-encoding corruption without data loss.
- **Cloud and deployment** — AWS infrastructure for the Digital Health Platform, and operation of the Mines Administration System on Kubernetes with Jenkins pipelines, nginx and Docker.
- **Root-cause analysis** — Diagnosis and resolution of production incidents including an SSO and Keycloak redirect loop, cross-service workflow status divergence and CORS misconfiguration.
- **Stakeholder engagement** — Requirement-gathering sessions and user acceptance testing with business users, translating process requirements into technical specifications and documentation.

5. TECHNICAL SKILLS

Languages	Java, JavaScript and TypeScript, Python, SQL, HTML, CSS
Backend	Spring Boot, Spring MVC, Spring Security, Hibernate and JPA, REST APIs, Node.js, FastAPI, Django
Frontend	Angular, Next.js and React
Databases	PostgreSQL, MySQL, MongoDB; backup and restoration, foreign data wrappers, indexing, view and query optimisation
Cloud and DevOps	AWS (Lambda, API Gateway, EC2, S3, IAM, Cognito, EventBridge, VPC, Secrets Manager), Kubernetes, Docker, Jenkins, nginx, Linux, Git, Maven
Digital identity	Bhutan NDI — credential issuance, proof requests, passwordless login, webhook registration, NATS event handling
Security	OWASP ZAP remediation, content security policy and headers, JWT, role-based access control, SSO with Keycloak, dependency scanning, patch management, secure API design
Practices	Requirement gathering, system analysis and design, technical specification, user acceptance testing, production support, project coordination

6. EDUCATION

Qualification	Institution	Year	Result
Bachelor of Technology (B.Tech) Computer Science & Engineering (Cloud Technology & Information Security)	M. M. Engineering College, Mullana Maharishi Markandeshwar (Deemed to be University), India	2019 – 2023	CGPA 8.38 Distinction
Bhutan Higher Secondary Education Certificate (Class XII, Science)	Samtse Higher Secondary School Bhutan Council for School Examinations and	2018	58.4%

	Assessment		
Bhutan Certificate of Secondary Education (Class X)	Samtse Higher Secondary School Bhutan Council for School Examinations and Assessment	2016	73.0%

7. CERTIFICATIONS AND HACKATHONS

Certification / Training / Award	Issuing Body	Date
AWS Certified Cloud Practitioner (CLF-C02), valid to Sep 2027	Amazon Web Services	Sep 2024
Certified DevOps Engineer (CDE)	Global Skill Development Council	Sep 2025
PMO and Solution Architecture Capacity Development Program	Accenture Japan and JICA	Feb 2025
Web Application and API Protection, Level 1	Prophaze Technologies	Aug 2023
Introduction to Data Science with Python	Omdena Academy	Dec 2024
Participant, Bhutan DevFest International Hackathon	GovTech Agency, Bhutan NDI and Ethereum Foundation	Jun 2025
Participant, Blockchain-Based Land Registry Hackathon	GovTech Agency and National Land Commission	Jun 2025
Participant, DHI and AWS Hackathon	Druk Holding and Investments with AWS	Oct 2024

Referees available on request.